

6 CONCLUSION

Reformer combines the modeling capacity of a Transformer with an architecture that can be executed efficiently on long sequences and with small memory use even for models with a large number of layers. We believe that this will help large, richly-parameterized Transformer models become more widespread and accessible. Also, the ability to handle long sequences opens the way for the use of the Reformer on many generative tasks. In addition to generating very long coherent text, the Reformer can bring the power of Transformer models to other domains like time-series forecasting, music, image and video generation.

REFERENCES

- Rami Al-Rfou, Dokook Choe, Noah Constant, Mandy Guo, and Llion Jones. Character-level language modeling with deeper self-attention. *CoRR*, abs/1808.04444, 2018. URL <http://arxiv.org/abs/1808.04444>.
- Alexandr Andoni, Piotr Indyk, Thijs Laarhoven, Ilya P. Razenshteyn, and Ludwig Schmidt. Practical and optimal LSH for angular distance. *CoRR*, abs/1509.02897, 2015. URL <http://arxiv.org/abs/1509.02897>.
- Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016. URL <http://arxiv.org/abs/1607.06450>.
- Antoine Bordes, Nicolas Usunier, Sumit Chopra, and Jason Weston. Large-scale simple question answering with memory networks. *CoRR*, abs/1506.02075, 2015. URL <http://arxiv.org/abs/1506.02075>.
- Sarath Chandar, Sungjin Ahn, Hugo Larochelle, Pascal Vincent, Gerald Tesauro, and Yoshua Bengio. Hierarchical memory networks. *arXiv preprint arXiv:1605.07427*, 2016.
- Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. URL <https://openai.com/blog/sparse-transformers>, 2019.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. *CoRR*, abs/1810.04805, 2018. URL <http://arxiv.org/abs/1810.04805>.
- Aidan N Gomez, Mengye Ren, Raquel Urtasun, and Roger B Grosse. The reversible residual network: Backpropagation without storing activations. In *Advances in neural information processing systems*, pp. 2214–2224, 2017.
- Felix Hill, Antoine Bordes, Sumit Chopra, and Jason Weston. The goldilocks principle: Reading children’s books with explicit memory representations. *CoRR*, abs/1511.02301, 2015. URL <http://arxiv.org/abs/1511.02301>.
- Cheng-Zhi Anna Huang, Ashish Vaswani, Jakob Uszkoreit, Noam Shazeer, Curtis Hawthorne, Andrew M Dai, Matthew D Hoffman, and Douglas Eck. Music transformer: Generating music with long-term structure. *arXiv preprint arXiv:1809.04281*, 2018.
- Guillaume Lample, Alexandre Sablayrolles, Marc’Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. Large memory layers with product keys. *CoRR*, abs/1907.05242, 2019. URL <http://arxiv.org/abs/1907.05242>.
- Peter J. Liu, Mohammad Saleh, Etienne Pot, Ben Goodrich, Ryan Sepassi, Lukasz Kaiser, and Noam Shazeer. Generating wikipedia by summarizing long sequences. *CoRR*, abs/1801.10198, 2018. URL <http://arxiv.org/abs/1801.10198>.
- Myle Ott, Sergey Edunov, David Grangier, and Michael Auli. Scaling neural machine translation. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pp. 1–9, Brussels, Belgium, October 2018. Association for Computational Linguistics. doi: 10.18653/v1/W18-6301. URL <https://www.aclweb.org/anthology/W18-6301>.

- Niki Parmar, Ashish Vaswani, Jakob Uszkoreit, Lukasz Kaiser, Noam Shazeer, and Alexander Ku. Image transformer. *CoRR*, abs/1802.05751, 2018. URL <http://arxiv.org/abs/1802.05751>.
- Matt Post. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pp. 186–191, Belgium, Brussels, October 2018. Association for Computational Linguistics. URL <https://www.aclweb.org/anthology/W18-6319>.
- Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. 2019.
- Jack W Rae, Jonathan J Hunt, Tim Harley, Ivo Danihelka, Andrew Senior, Greg Wayne, Alex Graves, and Timothy P Lillicrap. Scaling memory-augmented neural networks with sparse reads and writes. In *Advances in Neural Information Processing Systems, (NIPS)*, 2016.
- Prajit Ramachandran, Niki Parmar, Ashish Vaswani, Irwan Bello, Anselm Levskaya, and Jonathon Shlens. Stand-alone self-attention in vision models. *CoRR*, abs/1906.05909, 2019. URL <http://arxiv.org/abs/1906.05909>.
- Adam Santoro, Sergey Bartunov, Matthew Botvinick, Daan Wierstra, and Timothy P. Lillicrap. One-shot learning with memory-augmented neural networks. *CoRR*, abs/1605.06065, 2016. URL <http://arxiv.org/abs/1605.06065>.
- Noam Shazeer and Mitchell Stern. Adafactor: Adaptive learning rates with sublinear memory cost. *CoRR*, abs/1804.04235, 2018. URL <http://arxiv.org/abs/1804.04235>.
- Noam Shazeer, Youlong Cheng, Niki Parmar, Dustin Tran, Ashish Vaswani, Penporn Koanantakool, Peter Hawkins, HyukJoong Lee, Mingsheng Hong, Cliff Young, Ryan Sepassi, and Blake Hechtman. Mesh-tensorflow: Deep learning for supercomputers. *CoRR*, abs/1811.02084, 2018. URL <http://arxiv.org/abs/1811.02084>.
- Nimit Sharad Sohoni, Christopher Richard Aberger, Megan Leszczynski, Jian Zhang, and Christopher Ré. Low-memory neural network training: A technical report. *CoRR*, abs/1904.10631, 2019. URL <http://arxiv.org/abs/1904.10631>.
- Sainbayar Sukhbaatar, Edouard Grave, Piotr Bojanowski, and Armand Joulin. Adaptive attention span in transformers. *CoRR*, abs/1905.07799, 2019a. URL <http://arxiv.org/abs/1905.07799>.
- Sainbayar Sukhbaatar, Edouard Grave, Guillaume Lample, Hervé Jégou, and Armand Joulin. Augmenting self-attention with persistent memory. *CoRR*, abs/1907.01470, 2019b. URL <http://arxiv.org/abs/1907.01470>.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. *CoRR*, 2017. URL <http://arxiv.org/abs/1706.03762>.
- Jason Weston, Sumit Chopra, and Antoine Bordes. Memory networks. *CoRR*, abs/1410.3916, 2014. URL <http://arxiv.org/abs/1410.3916>.

A MULTI-ROUND LSH ATTENTION

In this section we describe in more detail the multi-hash version of our LSH attention mechanism. We first repeat Equation (3) from the main text, which describes a general formulation of attention with sparsity:

$$o_i = \sum_{j \in \tilde{\mathcal{P}}_i} \exp(q_i \cdot k_j - m(j, \mathcal{P}_i) - z(i, \mathcal{P}_i)) v_j \quad \text{where } m(j, \mathcal{P}_i) = \begin{cases} \infty & \text{if } j \notin \mathcal{P}_i \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

In the multi-round case, a query position i can attend to key positions $\mathcal{P}_i$ as defined in (6), which we also repeat here:

$$\mathcal{P}_i = \bigcup_{r=1}^{n_{\text{rounds}}} \mathcal{P}_i^{(r)} \quad \text{where } \mathcal{P}_i^{(r)} = \left\{ j : h^{(r)}(q_i) = h^{(r)}(q_j) \right\} \quad (6)$$

For batching purposes, attention is performed on chunks of sorted queries/keys:

$$\tilde{\mathcal{P}}_i^{(r)} = \left\{ j : \left\lfloor \frac{s_i^{(r)}}{m} \right\rfloor - 1 \leq \left\lfloor \frac{s_j^{(r)}}{m} \right\rfloor \leq \left\lfloor \frac{s_i^{(r)}}{m} \right\rfloor \right\} \quad (11)$$

Combining (3) and (6) gives:

$$o_i = \sum_{j \in \tilde{\mathcal{P}}_i} \exp(q_i \cdot k_j - m(j, \mathcal{P}_i) - z(i, \mathcal{P}_i)) v_j \quad (12)$$

$$= \sum_{r=1}^{n_{\text{rounds}}} \exp(z(i, \mathcal{P}_i^{(r)}) - z(i, \mathcal{P}_i)) \sum_{j \in \tilde{\mathcal{P}}_i^{(r)}} \frac{1}{N_{i,j}} \exp(q_i \cdot k_j - m(j, \mathcal{P}_i^{(r)}) - z(i, \mathcal{P}_i^{(r)})) v_j \quad (13)$$

$$= \sum_{r=1}^{n_{\text{rounds}}} \exp(z(i, \mathcal{P}_i^{(r)}) - z(i, \mathcal{P}_i)) o_i^{(r)} \quad (14)$$

$$o_i^{(r)} = \sum_{j \in \tilde{\mathcal{P}}_i^{(r)}} \exp(q_i \cdot k_j - m_{i,j}^{(r)} - z(i, \mathcal{P}_i^{(r)})) v_j \quad (15)$$

$$\text{where } N_{i,j} = \left| \left\{ r' : j \in \mathcal{P}_i^{(r')} \right\} \right| \text{ and } m_{i,j}^{(r)} = \begin{cases} \infty & \text{if } j \notin \mathcal{P}_i^{(r)} \\ 10^5 & \text{if } i = j \\ \log N_{i,j} & \text{otherwise} \end{cases} \quad (16)$$

Each round of LSH attention produces a vector $o_i^{(r)}$ that can be computed independently from other rounds, except for the inclusion of a term $N_{i,j}$ to avoid double-counting elements when constructing the union of $\mathcal{P}_i^{(r)}$ sets. In our implementation we fold the $N_{i,j}$ factor into the masking term $m_{i,j}^{(r)}$.

We also modify $m_{i,j}^{(r)}$ to introduce a special case for $i = j$. This case is added because causal masking in a standard Transformer allows position i to attend to itself, which is not desirable in a shared-QK formulation. We set the mask to a large but finite value to disallow attention-in-place, except in the situation where a token has no other valid attention targets. For example, the first token in a sequence attends only to itself, because no prior context is available.